



Teacher Copy — includes the Answer Key & Teacher Guide. Do not distribute to students.

UNIT 2 • STANDARD 6.NOS.1

Interpret Division of Fractions

Flagship

Lesson 2-1-flagship · Teacher Copy

Name: _____

Date: _____

Class: _____

EVIDENCE LAB MISSION

The Fraction Vault

You are the lead forensic analyst at the Reyes Detective Agency. A 3-foot strip of evidence tape must be cut into exact $\frac{1}{4}$ -foot sections before it can be logged — and the vault that holds the case file only opens once you know how many pieces that makes. Interpret fraction division and crack the case.



TODAY'S OBJECTIVES

Content Objective: I can use models to interpret what it means to divide by a fraction.

Language Objective: I can explain a fraction division using the words dividend, divisor, quotient, reciprocal, and unit fraction.



See **Learn It** for the explanation and worked examples, and the **Vocab** tab for the words. These are your notes to fill in and keep.



Fill in each blank as we go. Use the Word Bank to help you.



WORD BANK — FILL EACH BLANK WITH THE BEST WORD

Dividend

Divisor

Quotient

Reciprocal

Unit fraction



Tap any word to see what it means and a picture.

1 In $6 \div 1/2$, the number 6 being divided is the .

2 In $6 \div 1/2$, the number $1/2$ we divide by is the .

3 The answer to a division problem is the .

4 The fraction you get by flipping the numerator and denominator is the .

5 A fraction with a numerator of 1, like $1/5$, is a .

Write About the Math

Guided ESOL writing

Use the support level you need. Every level answers the same math question.

1. Understand the Question

EXPLAIN

**On your fraction bar, how many $\frac{1}{4}$ -foot sections fit into one whole foot?
Point to where you see the pieces?**

Your job: Answer the question. Use math evidence. Explain how the evidence proves your answer.

Responde la pregunta. Usa evidencia matemática. Explica cómo la evidencia demuestra tu respuesta.

2. Plan Your Math Words

Check at least two words you will use.

☐ **Dividend** — The number you are splitting up in a division problem.
Dividendo — El número que estás repartiendo en una división.

☐ **Divisor** — The number you split by in a division problem.
Divisor — El número entre el que divides en una división.

☐ **Quotient** — The answer when you divide.
Cociente — La respuesta cuando divides.

☐ **Unit fraction** — A fraction with 1 on top, like $\frac{1}{4}$.
Fracción unitaria — Una fracción con 1 arriba, como $\frac{1}{4}$.

☐ **Reciprocal** — A fraction turned upside down.
Recíproco — Una fracción volteada de arriba abajo.

Say it first: Say your idea to a partner or quietly to yourself before you write.

Di tu idea a un compañero o en voz baja antes de escribir.

In one whole foot I see ____ sections of $\frac{1}{4}$ each.

En un pie completo veo ____ secciones de $\frac{1}{4}$ cada una.

3. Build Your Explanation

Model the parts: This model shows the parts of an explanation. It does not answer your writing question.

Claim: Because each piece is smaller than a whole foot, 3 feet makes more than 3 pieces.

Evidence: Students reason that since $\frac{1}{4}$ is less than 1, more than 3 pieces fit, so the quotient is greater than the dividend.

Reasoning: This evidence connects the definition of dividend to the mathematical claim.

Start

Most language support

Complete the frames. Use the word bank.

Completa las oraciones. Usa el banco de palabras.

- **In one whole foot I see ____ sections of $\frac{1}{4}$ each.**

En un pie completo veo ____ secciones de $\frac{1}{4}$ cada una.

- **So 3 feet hold ____ pieces because 3 times ____ equals ____.**

Entonces 3 pies tienen ____ pedazos porque 3 por ____ es ____.

Build

Some language support

Write two connected sentences. Use because, so, or but.

Escribe dos oraciones conectadas. Usa porque, entonces o pero.

- **My claim is ____.**

Mi afirmación es ____.

- **I know because ____.**

Lo sé porque ____.

- **So ____.**

Entonces ____.

Explain

Light language support

Write a claim, give math evidence, and explain your reasoning.

Escribe una afirmación, da evidencia matemática y explica tu razonamiento.

- **My claim is ____.**
Mi afirmación es ____.
- **My math evidence is ____.**
Mi evidencia matemática es ____.
- **This proves ____ because ____.**
Esto demuestra ____ porque ____.

4. Check Your Explanation

- ☐ I answered the question.
Respondí la pregunta.

- ☐ I used math evidence: numbers, an equation, a model, or a comparison.
Usé evidencia matemática: números, una ecuación, un modelo o una comparación.

- ☐ I explained how the evidence proves my answer.
Explicué cómo la evidencia demuestra mi respuesta.

- ☐ I used at least two math words.
Usé por lo menos dos palabras matemáticas.

- ☐ I reread complete sentences and punctuation.
Volví a leer mis oraciones completas y la puntuación.

Writing Guide — Teacher Copy

The support level changes the amount of language scaffolding, not the mathematical expectation. Look for the same five features in every response.

- I answered the question.
- I used math evidence: numbers, an equation, a model, or a comparison.
- I explained how the evidence proves my answer.
- I used at least two math words.
- I reread complete sentences and punctuation.

Answer Key & Teacher Guide

1. **Notes 1:** Dividend
2. **Notes 2:** Divisor
3. **Notes 3:** Quotient
4. **Notes 4:** Reciprocal
5. **Notes 5:** Unit fraction
6. **Watch for this mistake:** *A common mistake in Interpret Division of Fractions is assuming division always makes the answer smaller — a rule that holds for whole numbers but breaks down when dividing by a fraction less than 1. For example, in $6 \div 1/2$, a student expects a smaller answer and calculates $6 \times 1/2 = 3$, when the correct answer is $6 \div 1/2 = 12$, because 12 halves fit inside 6 wholes. The same error shows up in word problems: if a recipe needs $1/2$ cup of flour per batch and you have 6 cups, a student might reason 'dividing means less' and answer 3 batches instead of the correct 12 batches. Before you submit, ask: am I dividing by a fraction smaller than 1? If so, the answer should be BIGGER than the number I started with.*
7. **Try It 1:** A. $8 \div 1/4 = 32$ sections — *Each foot holds 4 fourths, so 8 feet hold $8 \times 4 = 32$ sections. The matching expression is $8 \div 1/4 = 32$.*
8. **Try It 2:** A. The number of $1/3$ -quart beakers she can fill, which is 15 — *$5 \div 1/3$ asks how many $1/3$ -quart beakers fit into 5 quarts. Each quart holds 3 thirds, so $5 \times 3 = 15$ beakers.*